

Business Context & Problem Definition

A retail organization aims to gain deeper insights into **customer purchasing behavior** to support revenue growth, improve customer experience, and strengthen long-term retention. Recent shifts in consumer behavior across **demographics, product categories, and purchasing patterns** have created the need for a data-driven understanding of what influences buying decisions.

The business seeks to analyze how factors such as **discounts, product ratings, seasonal trends, subscription status, and purchase frequency** impact customer engagement and overall value.

Core Business Question

How can customer shopping data be used to uncover meaningful patterns, identify high-value customer segments, and enhance marketing and product strategies through data-driven insights?

Project Scope & Deliverables

- 1. Data Cleaning & Feature Engineering (Python)**
Prepare and transform raw customer data by handling inconsistencies, creating new analytical features, and performing exploratory analysis.
- 2. Business Analysis & Querying (SQL)**
Structure the dataset into a relational format and execute SQL queries to evaluate customer segmentation, loyalty trends, revenue drivers, and subscription impact.
- 3. Dashboard Development & Visualization (Power BI)**
Design an interactive dashboard that communicates key insights related to customer value, revenue distribution, and purchasing trends for business stakeholders.
- 4. Insights & Recommendations Report**
Summarize analytical findings and translate them into actionable business recommendations supported by visual evidence.
- 5. Version-Controlled Project Repository (GitHub)**
Maintain a clean and organized GitHub repository containing Python scripts, SQL queries, Power BI files, and documentation.